Date:14/10/26

Experiment – 4

Verification of System Properties: Linearity, Time Invariance, Causality and Stability

OBJECTIVE:

i) To verify the linearity and time-invariance properties of continuous-time and discrete-time systems using suitable input signals and system responses.

SOFTWARE / LABORATORY ENVIRONMENT:

MATLAB

I. To verify the linearity and time-invariance properties of continuous-time and discrete-time systems using suitable input signals and system responses.

PROCEDURE:

1. Select a representative CT or DT system.
2. Generate two independent test inputs.
3. Compute the individual system outputs.
4. Form a combined input and compare its system output with the corresponding linear
5. combination of outputs.
6. Delay the input by t0 and compare the system response with the correspondingly
7. delayed original output.
8. Calculate numerical differences and classify the system**.**

MATLAB CODE:

```
clc; clear; close all;

t = -1:0.001:1;

x1 = sin(2*pi*5*t); x2 = cos(2*pi*5*t);

a = 2; b = 3; t0 = 0.2;

T = @(x) 2*x;

lhs = T(a*x1+b*x2);
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```
rhs = a*T(x1)+b*T(x2);

linError = max(abs(lhs-rhs));

xDelay = sin(2*pi*5*(t-t0));

lhsTI = T(xDelay);

rhsTI = T(x1);

rhsTIshift = interp1(t,rhsTI,t-t0,'linear',0);

tiError = max(abs(lhsTI-rhsTIshift));

figure;

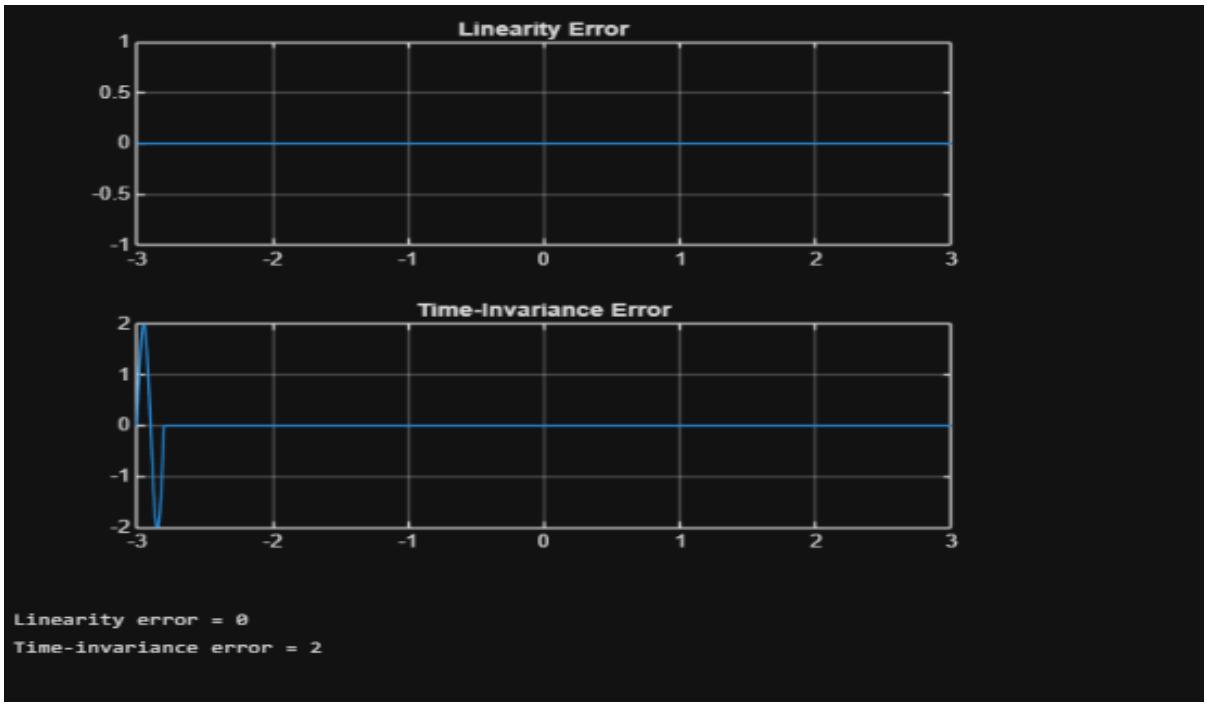
subplot(2,1,1); plot(t,lhs-rhs); grid on; title('Linearity Error');

subplot(2,1,2); plot(t,lhsTI-rhsTIshift); grid on; title('Time-Invariance Error');

fprintf('Linearity error = %g\n',linError);

fprintf('Time-invariance error = %g\n',tiError);
```

OUTPUT:



RESULT:

Linearity and Time-invariance properties of continuous-time and discrete-time systems is successfully verified.

LABORATORY PERFORMANCE ASSESSMENT RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks given	Marks obtained
(i) Determine the inverse Laplace transform of a given frequency-domain representation and reconstruct the corresponding time-domain signal.	**Inverse Laplace Transform Evaluation**	Correctly expands X(s) using partial fractions (distinct, repeated and complex poles) and applies the standard transform pairs and properties. Obtains the exact time-domain expression x(t), including the unit step u(t) and any shifts, with correct constants.	Applies partial fractions and transform pairs correctly, with minor errors in residues, constants or step-function terms.	Performs basic inverse transforms but has noticeable errors in partial fractions or transform pairs, and needs guidance.	Unable to determine the inverse Laplace transform.	**20**	**2**	
	Time-Domain Reconstruction & MATLAB Implementation	Uses suitable MATLAB commands (residue, ilaplace, syms, impulse or step) with a proper time vector. Reconstructs and plots x(t) with titles, axis labels and grid, and verifies the numerical result against the analytical expression.	Reconstructs and plots the signal correctly, with minor errors in code, time range, parameters or labelling.	Generates a basic reconstruction with noticeable errors in code or plots, and needs assistance.	Unable to reconstruct the time-domain signal or produce meaningful plots.	**20**	**2**	

(ii) Determine the Region of Convergence (ROC) of Laplace transforms and analyze its significance in characterizing continuous-time signals and systems.	**ROC Determination**	Correctly determines the ROC for right-sided, left-sided, two-sided and finite-duration signals from the pole locations. States the ROC as a proper inequality on Re(s) and shows it on the s-plane with poles and zeros marked.	Determines the ROC correctly for most signals, with minor errors in the boundary, signal type or s-plane plot.	Determines the ROC for basic cases only, with noticeable errors, and needs guidance.	Unable to determine the ROC of the given Laplace transforms.	**20**	**2**	
	ROC Significance for Signals and Systems	Clearly relates the ROC to the signal type, causality and BIBO stability (ROC contains the jω-axis), and shows that the same X(s) with different ROCs gives different time signals. Supports the reasoning with mathematical conditions and pole-zero evidence.	Relates the ROC to causality and stability correctly, with minor errors or limited justification.	Gives a basic explanation of the ROC's significance. Justification is incomplete and needs guidance.	Unable to explain the significance of the ROC.	**20**	**2**	
Interpret the Laplace-domain results and document the experiment.	**Engineering Analysis, Interpretation & Record**	Provides technically accurate interpretation of the results (pole locations, ROC and signal behaviour). Maintains a complete, well-organized record with code, outputs, plots and	Maintains a proper record with generally accurate interpretation, with minor omissions.	Record and interpretation are basic and incomplete, with some technical inaccuracies.	Incomplete record, limited or incorrect interpretation of the results.	**20**	**2**	

		conclusions.						
					Total	**100**	**10**	